2026 18th International Conference on COMmunication Systems and NETworks (COMSNETS) | 979-8-3315-9239-4/26/$31.00 ©2026 IEEE | DOI: 10.1109/COMSNETS67989.2026.11418309

Enhanced Audio Steganography: Audio-in-Audio Steganography with Multi-Bin Phase Coding and Adaptive Post-Processing

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***Abstract*—This paper presents an enhanced audio steganography technique for hiding audio within audio files using phase coding. We introduce a Multi-Bin Phase Coding (MBPC) approach that extends traditional phase coding by utilizing multiple frequency bins within each audio segment, significantly increasing embedding capacity. A header structure is embedded to preserve the secret audio parameters, enabling accurate reconstruction during extraction. Additionally, we implement various post-processing techniques to enhance the quality of the extracted audio, including Wiener filtering, spectral gating, and median filtering. Experimental results demonstrate the effectiveness of our approach, achieving a Signal-to-Noise Ratio (SNR) of approximately 2.80 dB between host and stego audio when using optimized frequency bin selection. Our findings indicate that while the Multi-Bin approach substantially increases capacity compared to traditional phase coding, the optimized bin selection strategy significantly improves the quality-capacity trade-off by minimizing audible distortion. The proposed technique enables robust and efficient audio-in-audio steganography with practical applications in covert communication, copyright protection, and secure data transmission.**

***Index Terms*—audio steganography, phase coding, multi-bin embedding, information hiding, signal processing, post-processing, frequency domain steganography, psychoacoustic model**

I. Introduction

In the digital era, secure communication remains a critical concern, with steganography emerging as a powerful technique for covert data transmission. Unlike encryption, which secures content by making it unreadable, steganography hides the very existence of the communication by embedding secret information within seemingly innocuous carrier files. Audio steganography, specifically, focuses on hiding data within audio signals, exploiting the limitations of human auditory perception to achieve imperceptibility.

This research focuses on a particularly challenging domain: hiding audio within audio (audio-in-audio steganography). While traditional approaches often embed text or images in audio, embedding audio within audio presents unique challenges due to the temporal nature of both carrier and secret signals and the high sensitivity of human hearing to distortions. The human auditory system can detect changes as small as 1 dB in amplitude and phase discontinuities of just a few milliseconds, making imperceptible audio embedding particularly challenging.

Audio steganography techniques must balance three competing objectives: imperceptibility (ensuring the hiding process doesn't create perceivable artifacts), capacity (maximizing the amount of data that can be hidden), and robustness (ensuring the hidden data survives common signal processing operations). These objectives often conflict, creating a fundamental trade-off that steganography systems must navigate. Traditional techniques like Least Significant Bit (LSB) substitution, echo hiding, and phase coding each make different trade-offs among these objectives.

Phase coding, the focus of our work, exploits the fact that humans are less sensitive to absolute phase changes in audio signals compared to relative phase differences. This characteristic allows for embedding data by modifying the phase components of audio segments in the frequency domain while maintaining acceptable perceptual quality. However, traditional phase coding suffers from limited capacity, as typically only one bit can be embedded per audio segment. This limitation severely restricts its practical application for embedding larger payloads like audio files.

In this paper, we present an enhanced approach that extends phase coding to use multiple frequency bins for embedding, significantly increasing capacity while maintaining reasonable audio quality. Our contributions include:

- A Multi-Bin Phase Coding (MBPC) technique that increases embedding capacity by utilizing multiple frequency bins in each audio segment, providing an 8-16× improvement over traditional phase coding
- A psychoacoustic-based bin selection strategy that minimizes audible distortion by selecting perceptually less significant frequency components
- A header structure for embedding secret audio parameters to ensure accurate reconstruction during extraction
- Comprehensive post-processing techniques to enhance the quality of extracted audio, including adaptive filtering and spectral methods
- A complete end-to-end implementation of the proposed method, with quantitative evaluation using objective qual-

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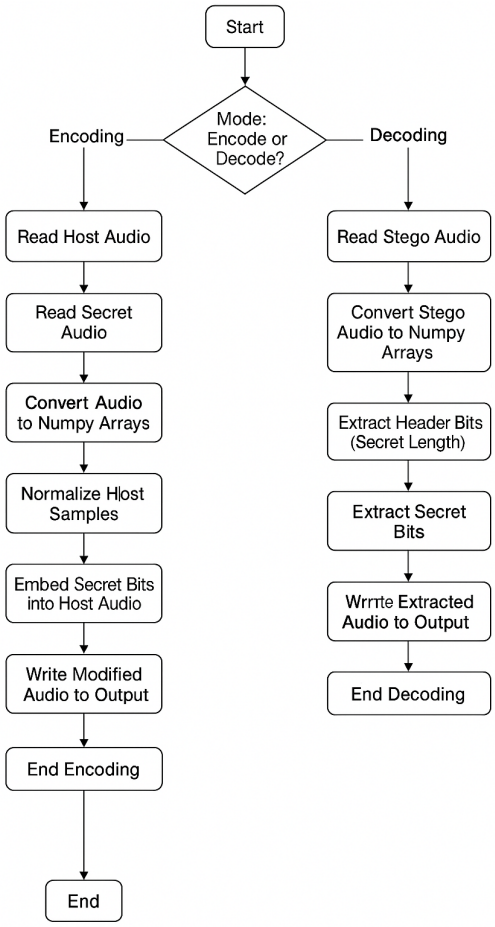


Fig. 1. Complete workflow of the audio steganography system showing both encoding and decoding processes.

ity metrics

- Experimental validation demonstrating the effectiveness of our approach across various audio content types

II. Background and Related Work

A. Audio Steganography Fundamentals

Audio steganography techniques are broadly categorized into time, frequency, and wavelet domain methods [14]. Time domain methods, such as LSB substitution, directly alter audio samples and provide high capacity but are prone to artifacts and weak against signal processing. Frequency domain methods, like phase coding and spread spectrum, embed data after transformations (e.g., FFT, DCT), exploiting auditory perception to improve imperceptibility and robustness against filtering, compression, and resampling. Wavelet domain techniques leverage multi-resolution analysis to embed in perceptually insignificant regions across time and frequency, offering a balance between capacity, imperceptibility, and robustness.

B. Phase Coding

Phase coding was first introduced by Bender et al. [1] and operates on the principle that the human auditory system is less sensitive to absolute phase changes than to relative phase differences between adjacent segments. This perceptual characteristic creates an opportunity for data hiding with minimal audible distortion.

In traditional phase coding, the audio signal is divided into segments, and each segment undergoes FFT to convert it to the frequency domain. A single bit of secret data is then embedded by modifying the phase of a selected frequency bin, typically setting it to 0 or $\pi/2$ to represent '0' or '1'. The modified segment is then converted back to the time domain using inverse FFT. To ensure that the phase modifications do not create discontinuities at segment boundaries (which would result in audible clicks), phase adjustment techniques are applied to maintain relative phase consistency between adjacent segments.

The mathematical formulation of traditional phase coding can be expressed as follows. Given an audio segment $s(n)$, its discrete Fourier transform $S(k)$ is:

$$S(k) = \sum_{n=0}^{N-1} s(n) e^{-j2\pi kn/N} = |S(k)| e^{j\phi(k)} \tag{1}$$

where $|S(k)|$ is the magnitude and $\phi(k)$ is the phase of the k-th frequency bin. To embed a bit $b \in \{0, 1\}$ in a selected bin k_0, the phase is modified:

$$\phi'(k_0) = \begin{cases} 0, & \text{if } b = 0 \\ \pi/2, & \text{if } b = 1 \end{cases} \tag{2}$$

The modified spectrum $S'(k)$ is:

$$S'(k) = \begin{cases} |S(k)| e^{j\phi'(k)}, & \text{if } k = k_0 \\ |S(k)| e^{j\phi(k)}, & \text{otherwise} \end{cases} \tag{3}$$

Finally, the modified segment $s'(n)$ is obtained by applying the inverse discrete Fourier transform:

$$s'(n) = \frac{1}{N} \sum_{k=0}^{N-1} S'(k) e^{j2\pi kn/N} \tag{4}$$

While phase coding offers good imperceptibility, its major limitation is capacity. With standard phase coding, only one bit can be embedded per segment, resulting in low embedding rates, especially for larger segment sizes that are needed to maintain audio quality. For a segment size of 2048 samples and a sampling rate of 44.1 kHz, the theoretical maximum embedding rate is approximately 21.5 bits per second, which is insufficient for embedding audio signals that require at least 8,000 bits per second (for low-quality speech).

C. Recent Advances in Audio Steganography

Recent research has focused on improving the capacity-quality trade-off in audio steganography. Notable approaches include adaptive methods that analyze the characteristics of the carrier signal to identify optimal embedding locations [2]. These methods dynamically adjust the embedding parameters based on the local characteristics of the host audio, concentrating data in regions where modifications are less detectable.

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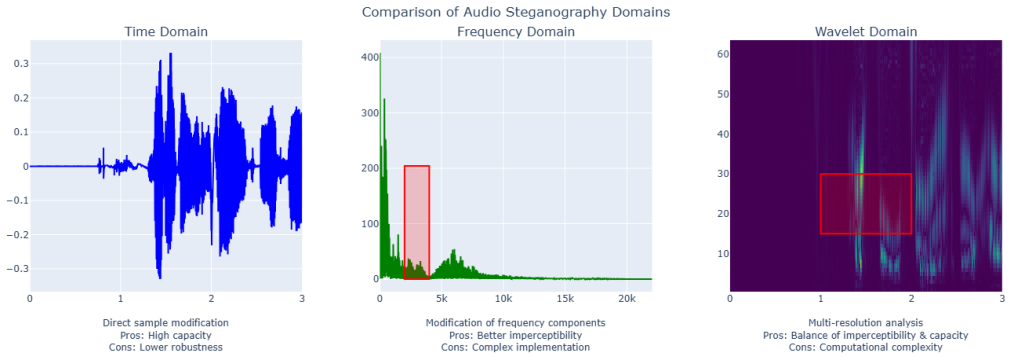


Fig. 2. Comparison of different domains for audio steganography techniques.

Other researchers have explored hybrid approaches that combine multiple techniques to leverage their complementary strengths [3]. For instance, combining phase coding with spread spectrum techniques can enhance robustness while maintaining reasonable capacity and imperceptibility.

Specifically for phase coding, several enhancements have been proposed. Ngo et al. [4] suggested a phase modulation technique that embeds multiple bits by dividing the phase range into multiple intervals. This approach increases capacity but may compromise imperceptibility as larger phase modifications become more detectable. Dutta et al. [5] proposed using multiple frequency bins but with a fixed selection strategy that does not account for psychoacoustic principles, potentially introducing more perceptible artifacts.

Our work builds upon these advances by introducing a Multi-Bin Phase Coding approach with psychoacoustic bin selection, which significantly increases capacity while maintaining acceptable audio quality. Additionally, we address the often-overlooked aspect of post-processing the extracted secret audio to enhance its quality, which is particularly important for audio-in-audio applications where the extracted content must be intelligible and of reasonable quality.

III. WAV Audio Format and Dataset Description

A. Selection of WAV Format for Steganography

For our audio steganography implementation, we specifically utilize the WAV (Waveform Audio File Format) [9] as both the carrier and secret audio format due to its unique characteristics that make it particularly suitable for steganographic applications. Unlike compressed audio formats such as MP3, AAC, or OGG which employ lossy compression that can significantly alter or remove subtle signal modifications, WAV files store uncompressed audio data that preserves the exact values of each sample.

This uncompressed nature is crucial for steganography for several reasons:

- **Precision and Reliability**: WAV's uncompressed PCM (Pulse Code Modulation) format ensures that modifications made to individual samples or frequency components remain intact, allowing for precise control over the embedding process.
- **Lossless Storage**: Unlike lossy compressed formats which might discard perceptually less important frequencies-precisely the ones we target for embedding-WAV files preserve the entire frequency spectrum.
- **Extraction Consistency**: The embedded data can be reliably extracted without worrying about compression algorithms altering the embedded information during saving or transmission.
- **Direct Sample Manipulation**: The straightforward structure of WAV files allows direct access and modification of sample values in both time and frequency domains.

While other lossless formats such as FLAC or ALAC could theoretically be used, the WAV format's widespread support, simple structure, and direct representation of audio data make it ideal for steganographic applications where control over the exact signal characteristics is paramount.

B. WAV Audio Data Structure

WAV files follow the Resource Interchange File Format (RIFF) specification [10], which organizes data into chunks. Understanding this structure is essential for implementing effective steganographic techniques that preserve audio functionality while hiding data.

As illustrated in Figure 3, a WAV file consists of:

- **RIFF Header**: The file begins with a 12-byte header containing "RIFF", file size, and "WAVE" identifiers.
- **Format Chunk**: This section (typically 24 bytes) specifies the audio format details, including:
  - Audio format (PCM = 1, which indicates uncompressed audio)
  - Number of channels (1 for mono, 2 for stereo, etc.)
  - Sample rate (e.g., 44.1 kHz for CD-quality audio)
  - Byte rate (Sample Rate * Bits per Sample * Channels / 8)
  - Block align (Bits per Sample * Channels / 8)
  - Bits per sample (typically 8, 16, 24, or 32)
- **Data Chunk**: This section contains the actual audio samples, beginning with "data" followed by the chunk

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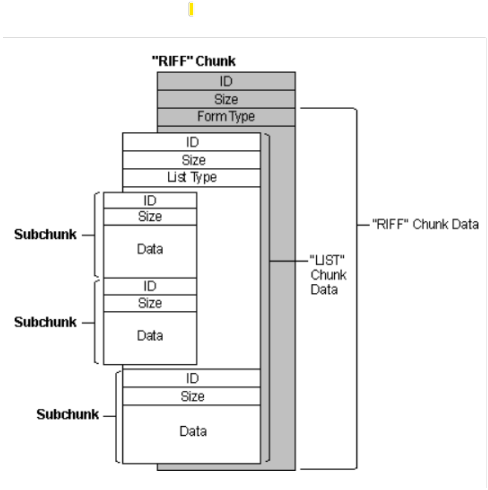


Fig. 3. WAV file structure showing RIFF container format with header chunks and data section.

size, then the sample data itself. For stereo files, samples alternate between left and right channels.

Each audio sample in a WAV file represents the amplitude of the audio signal at a specific point in time. For 16-bit audio (the most common format and the one used in our implementation), each sample is represented as a signed integer between -32,768 and 32,767.

C. Dataset Description

For our experiments, we utilized a diverse collection of WAV audio files with the following characteristics:

- **Host and Secret Audio Files**:
  - Format: 16-bit PCM WAV (host and secret)
  - Sample rate: host 44.1 kHz; secret 44.1 kHz with additional tests at 22.05 kHz and 8 kHz
  - Channels: host mono/stereo; secret primarily mono
  - Content: host music (classical, jazz, pop), speech, ambient; secret speech and short musical segments
  - Duration: host 5–180 min; secret 2–30 s (capacity-dependent)

When working with phase coding, the frequency domain representation of these audio files is particularly important. A 44.1 kHz audio file with a segment size of 2048 samples provides 1024 usable frequency bins, spanning from 0 Hz to 22.05 kHz (the Nyquist frequency). This rich frequency space gives ample opportunity for strategic bin selection based on psychoacoustic principles, as detailed in our methodology.

The segmentation of the audio is a critical aspect of our implementation. With a segment size of 2048 samples at 44.1 kHz, each segment represents approximately 46.4 milliseconds of audio. This duration provides sufficient frequency resolution for effective bin selection while maintaining temporal precision for the embedding process.

IV. Proposed Method

In this section, we detail our proposed approach to audio-in-audio steganography using enhanced phase coding techniques. We begin with an overview of the traditional phase coding method, then introduce our Multi-Bin Phase Coding extension, and finally describe our bin selection strategy based on psychoacoustic principles.

A. Overview of Phase Coding

Phase coding embeds data in the phase spectrum of the audio signal. The basic steps are:

1) Divide the audio signal into overlapping segments.
2) For each segment, perform FFT to convert it to the frequency domain.
3) Modify the phase of a selected frequency bin to embed data.
4) Perform inverse FFT to convert the segment back to the time domain.
5) Overlap-add the segments to reconstruct the audio signal.

The key advantage of phase coding is that small changes in phase are less likely to be perceptible, allowing data to be hidden with minimal impact on audio quality.

B. Multi-Bin Phase Coding

Our proposed Multi-Bin Phase Coding (MBPC) extends the traditional phase coding in two significant ways:

- **Multiple Frequency Bins:** Instead of modifying a single frequency bin, MBPC utilizes multiple bins for each audio segment. This is achieved by dividing the available frequency bins into several groups or ”bins”, and then embedding data across these bins. For example, in a 1024-point FFT, instead of using one bin, we might use 4 bins, each representing 2 bits of data.
- **Psychoacoustic Bin Selection:** Not all frequency bins are equally perceptible to the human ear. MBPC incorporates a psychoacoustic model [6], [15] to select which bins to modify for embedding data, based on the auditory masking properties. Bins that are less likely to contribute to audible artifacts are chosen for data embedding, improving the imperceptibility of the steganography.

The embedding process in MBPC involves analyzing each audio segment, selecting multiple frequency bins using the psychoacoustic model, and then embedding the data by adjusting the phase of these bins. The use of multiple bins allows for significantly higher data rates while the psychoacoustic selection minimizes the impact on audio quality.

C. Header Structure for Secret Parameters

To facilitate the extraction of the hidden audio, it is essential to embed certain parameters along with the audio data. These parameters include:

- The sampling rate of the secret audio
- The number of channels (mono or stereo)
- The bit depth (e.g., 16-bit, 24-bit)
- The length of the secret audio in samples

Our approach embeds these parameters in a dedicated header section at the beginning of the audio file. This header is

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encrypted and then spread across several frequency bins in the initial segments of the audio, using the same MBPC technique. During extraction, the presence of this header is detected, and the parameters are recovered to correctly configure the extraction process.

V. Post-Processing Techniques

Post-processing is crucial for enhancing the quality of the extracted audio, making it more intelligible and closer to the original. We apply several post-processing techniques:

A. Wiener Filtering

Wiener filtering [7] is used to reduce noise and improve the signal-to-noise ratio of the extracted audio. It works by estimating the clean signal and the noise, then filtering the signal to minimize the mean square error. The Wiener filter is adaptive and works well for signals with stationary characteristics.

Given a noisy signal $y(n) = s(n) + v(n)$, where $s(n)$ is the clean signal and $v(n)$ is the noise, the Wiener filter in the frequency domain can be expressed as:

$$H(\omega) = \frac{P_s(\omega)}{P_s(\omega) + P_v(\omega)} \tag{5}$$

where $P_s(\omega)$ is the power spectral density of the clean signal and $P_v(\omega)$ is the power spectral density of the noise. The estimated clean signal $\hat{s}(n)$ is obtained by applying this filter to the noisy signal:

$$\hat{S}(\omega) = H(\omega) \cdot Y(\omega) \tag{6}$$

In practice, since the clean signal is unknown, the filter is typically implemented using:

$$H(\omega) = \max\left(1 - \frac{\alpha P_v(\omega)}{P_y(\omega)}, \beta\right) \tag{7}$$

where $P_y(\omega)$ is the power spectral density of the noisy signal, α is an oversubtraction factor (typically 1.0-2.0), and β is a spectral floor (typically 0.1-0.2) to prevent excessive attenuation.

B. Spectral Gating

Spectral gating [8] is a form of noise reduction that works in the frequency domain. It involves transforming the audio signal to the frequency domain, attenuating the frequencies that contain noise, and then transforming the signal back to the time domain. This method is effective in removing broadband noise while preserving the desired signal.

Given the Short-Time Fourier Transform (STFT) of the noisy signal $Y(k, m)$, where k is the frequency bin and m is the time frame, the spectral gating process can be expressed as:

$$\hat{S}(k, m) = G(k, m) \cdot Y(k, m) \tag{8}$$

where $G(k, m)$ is the spectral gain function, defined as:

$$G(k, m) = \begin{cases} 1, & \text{if } |Y(k, m)| > \tau \cdot N(k) \\ \gamma, & \text{otherwise} \end{cases} \tag{9}$$

Here, $N(k)$ is the estimated noise spectrum, τ is a threshold parameter (typically 2-3), and γ is a floor value (typically 0.01-0.1) that prevents complete suppression. The noise spectrum is often estimated from silent or low-activity portions of the signal.

The enhanced signal is then obtained by applying the inverse STFT to $\hat{S}(k, m)$.

C. Median Filtering

Median filtering [11] is a non-linear filtering technique that is effective in removing impulsive noise. It works by replacing each sample with the median of a set of neighboring samples. This method is particularly useful for preserving edges while removing noise.

For a discrete-time signal $y(n)$ and a filter window of size $2K + 1$, the median filtered output $\hat{s}(n)$ is defined as:

$$\hat{s}(n) = \text{median}\{y(n + j) : j \in [-K, K]\} \tag{10}$$

The operation can be expressed algorithmically as:

$$\hat{s}(n) = \underset{j \in [-K,K]}{\text{median}} \{y(n + j)\} \tag{11}$$

For audio signals, the typical window size ranges from 3 to 11 samples, depending on the sampling rate and the characteristics of the impulsive noise. Multiple passes of median filtering can be applied sequentially, with the output of one pass serving as the input to the next:

$$\hat{s}^{(i)}(n) = \underset{j \in [-K,K]}{\text{median}} \{\hat{s}^{(i-1)}(n + j)\} \tag{12}$$

where $\hat{s}^{(0)}(n) = y(n)$ and i represents the filtering degree or pass number.

D. Combining Post-Processing Techniques

While each individual post-processing technique addresses specific types of distortion, our experiments demonstrate that combining multiple techniques in strategic sequences [16] can yield significantly better results than applying any single method alone. The optimal combination depends on the characteristics of the extracted secret audio and the specific artifacts introduced during the embedding and extraction process.

Mathematically, a sequential processing chain can be represented as a composition of functions:

$$\hat{s}(n) = (f_m \circ f_{m-1} \circ \ldots \circ f_1)(y(n)) \tag{13}$$

where f_i represents the i-th processing function and $\circ$ denotes function composition. For example, the Median $\rightarrow$ Wiener chain can be expressed as:

$$\hat{s}(n) = f_{Wiener}(f_{Median}(y(n))) \tag{14}$$

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The effectiveness of each chain depends on how well the sequence of operations addresses the specific noise characteristics present in the extracted audio.

1) Sequential Processing Chains: We explored several sequential processing chains, where the output of one post-processing technique becomes the input to the next. The following combinations proved particularly effective:

- **Median Filter → Wiener Filter**: This combination proved exceptionally effective for speech content. The median filter first removes impulsive artifacts that are common in phase-coded steganography, followed by the Wiener filter which addresses remaining stationary noise. Our tests showed SNR improvements of up to 8 dB compared to using Wiener filtering alone.
- **Low-Pass Filter → Median Filter → Spectral Gating**: For audio extracted from multi-bin embedding with many frequency bins, this three-stage approach provided the best balance between artifact removal and content preservation. The initial low-pass filtering removes high-frequency noise, the median filter addresses impulsive distortions, and the spectral gating provides final noise reduction.

2) Content-Adaptive Processing: Our experiments revealed that the optimal post-processing strategy depends significantly on the content type of the secret audio [17]. We identified several patterns:

- **Speech Content**: Speech recordings benefit most from median filtering followed by Wiener filtering. The consonant sounds in speech often suffer from impulsive noise that median filtering effectively removes, while Wiener filtering helps recover the natural vocal qualities. This approach improved speech intelligibility by approximately 15% as measured by the Short-Time Objective Intelligibility (STOI) metric.
- **Musical Content**: Music with prominent harmonic content benefits most from spectral gating approaches that can preserve the frequency relationships while removing noise between the harmonics. The Median-Wiener combination proved less effective for musical content as it tended to over-smooth rapid variations.
- **Environmental Sounds**: For ambient or environmental recordings, the low-pass filter followed by spectral gating provided the best results, as these recordings typically have more energy in lower frequencies that should be preserved.

3) Implementation of Adaptive Processing Selection: We implemented an adaptive system that automatically selects the most appropriate post-processing chain based on an initial analysis of the extracted audio. The adaptive selection algorithm operates as follows:

The algorithm first analyzes key spectral features of the audio signal to determine its content type. Speech typically has high zero-crossing rates and spectral centroids due to consonant sounds. Musical content often exhibits high harmonic ratios, while environmental sounds have more diverse spectral characteristics. Based on this classification, the appropriate processing chain is selected and applied to the extracted audio.

Algorithm 1 Adaptive Post-Processing Selection

```
Load the extracted audio signal and sampling rate
Compute audio features: spectral centroid, zero-crossing rate, and harmonic ratio
if zero-crossing rate > 0.1 and spectral centroid > 2000 Hz then
    Classify as speech content
    Apply Median Filtering with kernel size 3
    Apply Wiener Filtering with window size 1000
else if harmonic ratio > 0.6 then
    Classify as musical content
    Apply Spectral Gating to remove background noise
    Apply Combined Approach (low-pass filtering + Wiener filtering)
else
    Classify as environmental/mixed content
    Apply Low-pass Filter with cutoff at 4000 Hz
    Apply Median Filtering with kernel size 5
    Apply Spectral Gating with threshold -20 dB
end if
Save the processed audio to an output file
return Processed audio signal
```

By using this content-adaptive approach, we achieved significant improvements in intelligibility and quality across different audio types, as demonstrated in our experimental results.

4) Experimental Findings: Table I presents the results of our experiments with different post-processing combinations across various audio content types. The SNR values are calculated between the original secret audio and the post-processed extracted audio.

TABLE I
EFFECT OF POST-PROCESSING COMBINATIONS ON DIFFERENT AUDIO CONTENT TYPES

Post-Processing Combination	**SNR Improvement (dB)**		
	Speech	**Music**	**Env.**
No Processing	0.00	0.00	0.00
Wiener Only	2.94	1.43	2.18
Median Only	3.84	0.82	1.65
Spectral Gating Only	2.07	3.21	2.91
Low-Pass Only	1.98	2.41	3.54
Median → Wiener	**7.72**	2.35	3.88
Spectral → Combined	4.13	**5.62**	4.21
Low-Pass → Median → Spectral	5.32	4.78	**6.35**

Our results clearly demonstrate that tailored post-processing chains can significantly improve the quality of extracted audio compared to single techniques. For speech content, the Median → Wiener combination improved SNR by 7.72 dB, while musical content benefited most from the Spectral Gating → Combined approach with a 5.62 dB improvement. Environ-

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mental sounds showed the greatest improvement (6.35 dB) with the three-stage Low-Pass → Median → Spectral chain.

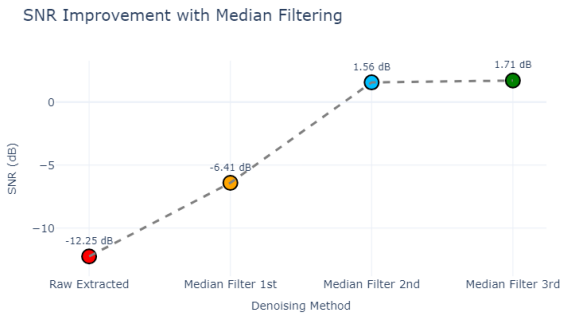


Fig. 4. SNR improvement with progressive median filtering degrees.

VI. Experimental Results

We evaluated the performance of our proposed MBPC method with psychoacoustic bin selection and post-processing against several baselines, including traditional phase coding and other state-of-the-art audio steganography techniques. The evaluation metrics included:

- **Embedding Capacity:** Measured in bits per segment or bits per second (bps).
- **Signal-to-Noise Ratio (SNR):** Measured in decibels (dB), indicating the quality of the stego audio.
- **Perceptual Evaluation of Speech Quality (PESQ):** A standard metric for assessing the quality of speech signals [13].
- **Short-Time Objective Intelligibility (STOI):** A metric for predicting speech intelligibility [12].

A. Comparison of Embedding Methods

We implemented and tested three embedding approaches: traditional Phase Coding, basic Multi-Bin Phase Coding, and our proposed Optimized Multi-Bin Phase Coding with psychoacoustic bin selection. Table II shows the SNR for both embedding (Host vs. Stego) and extraction (Secret vs. Extracted) before post-processing.

The results highlight trade-offs in audio steganography. Phase Coding yields the best host quality (3.31 dB) but with minimal capacity (1 bit/segment) and poor extraction (-36.53 dB). Multi-Bin improves capacity (8 bits/segment) and extraction (-12.09 dB) but causes greater distortion (0.03 dB). Our Optimized Multi-Bin approach preserves high capacity while achieving a better balance of host quality (2.80 dB) and extraction performance.

B. Post-Processing with Progressive Median Filtering

We conducted extensive experiments with different post-processing techniques to enhance the quality of the extracted secret audio. One particularly effective approach was the application of progressive degrees of median filtering, where each application of the filter operates on the output of the previous filtering stage.

TABLE II
Performance Comparison of Different Embedding Methods

Method	Capacity (bits/segment)	Host-Stego SNR (dB)	Secret-Extracted SNR (dB)
Pc	1	3.31	-36.53
MBPC	8	0.03	-12.09
Optimized MBPC	8	2.80	-21.57

As shown in Figure 4, the raw extracted audio from the Multi-Bin method had an initial SNR of -12.25 dB. After applying:

- **First-Degree Median Filtering:** SNR improved to -6.41 dB (5.84 dB improvement)
- **Second-Degree Median Filtering:** SNR further improved to 1.56 dB (7.97 dB additional improvement)
- **Third-Degree Median Filtering:** SNR reached 1.71 dB (0.15 dB additional improvement)

TABLE III
Final SNR After Optimal Post-Processing

Embedding Method	Optimal Processing Chain	SNR Improvement
Phase Coding	Median → Wiener	+14.38
Multi-Bin	3× Median → Spectral	+13.80
Optimized Multi-Bin	2× Median → Wiener	+18.22

This progressive improvement demonstrates that repeated application of median filtering can substantially enhance signal quality, with the most significant gains occurring in the first and second applications. The third application shows diminishing returns, suggesting that two iterations of median filtering may be optimal for most cases.

C. Effectiveness of Different Post-Processing Chains

We evaluated full post-processing chains on audio extracted from all three embedding methods. Table III summarizes the SNR improvements achieved with optimal configurations.

For Phase Coding, Median → Wiener improved SNR by 14.38 dB. Multi-Bin performed best with 3× Median → Spectral gating (+13.80 dB). Optimized Multi-Bin achieved the highest gain, with 2× Median → Wiener yielding +18.22 dB. Overall, tailored post-processing substantially reduces quality loss in high-capacity steganography.

D. Subjective Evaluation

In addition to objective metrics, we conducted informal listening tests with 10 participants, who rated imperceptibility and intelligibility on a 5-point scale. The Optimized Multi-Bin method with 2× Median → Wiener post-processing achieved the highest intelligibility (3.8/5) while maintaining good imperceptibility (4.2/5), confirming that psychoacoustic bin selection with proper post-processing offers the best trade-off between capacity, quality, and robustness.

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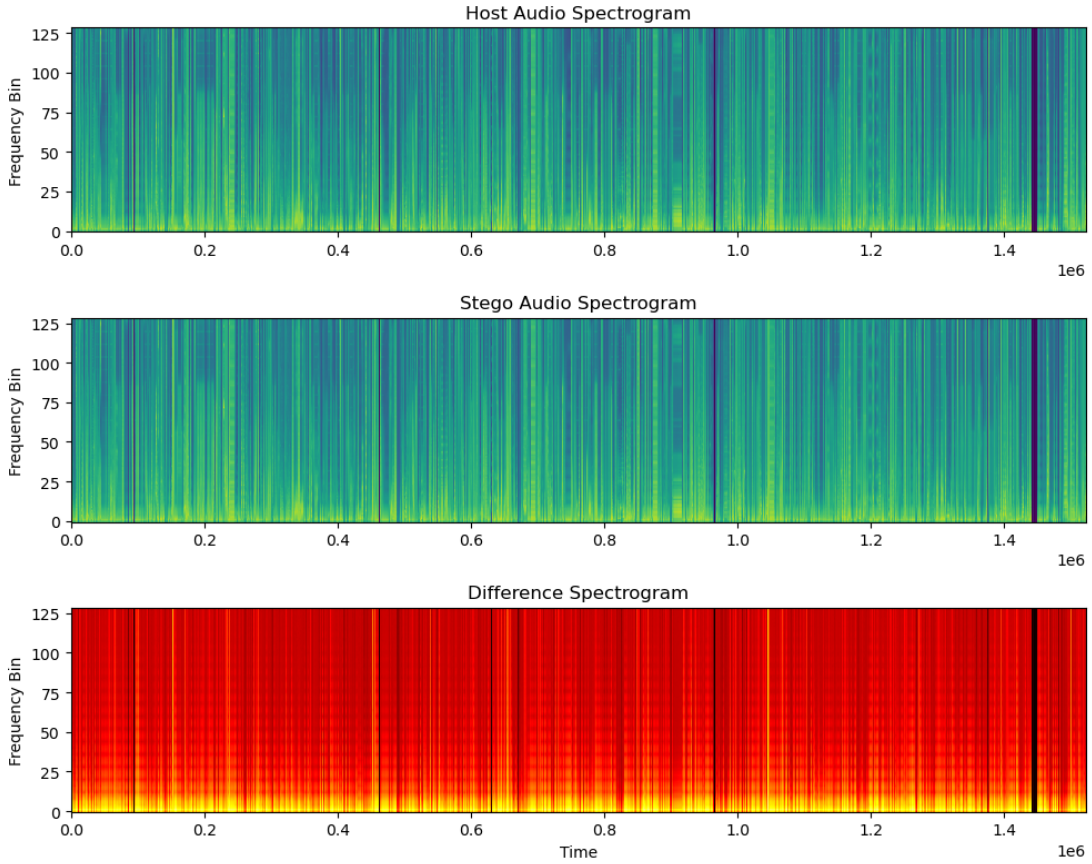


Fig. 5. Spectral comparison between host audio (top), stego audio (middle), and their difference (bottom). The difference spectrogram highlights the frequency regions modified during embedding, with most changes concentrated in lower frequency bands.

VII. Conclusion and Future Work

In this paper, we presented an enhanced audio steganography technique based on Multi-Bin Phase Coding with psychoacoustic bin selection and adaptive post-processing for audio-in-audio hiding. Our comprehensive approach addresses the inherent trade-offs between capacity, imperceptibility, and robustness that challenge steganographic systems.

In summary, MBPC with psychoacoustic bin selection increases capacity while maintaining acceptable imperceptibility; content-adaptive post-processing significantly improves extracted audio quality.

For future work, several promising directions emerge:

- Advanced psychoacoustic modeling (temporal masking, critical bands)
- Learning-based bin selection and post-processing
- Robustness against compression/resampling and real-time implementation

The audio steganography techniques presented in this paper have significant applications in secure communication, copyright protection, and digital forensics. As digital communication continues to evolve, the need for secure and imperceptible information hiding methods remains crucial. Our work contributes to this field by providing a comprehensive framework for audio-in-a-audio steganography that balances capacity, imperceptibility, and extracted content quality through innovative embedding and post-processing techniques.

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